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EDUCATION

B.S. in Aerospace Engineering, B.M. in Piano Performance **Aug 2022 – May 2027**
University of Florida, Gainesville, FL GPA: 3.52
Relevant: Nonlinear Optimization (Mini-Batch SGD), Microservices & API Gateways, Python (2+ Yrs), Sensor Data Pipelines, Numerical Simulation

TOP SKILLS

Programming | Modeling: Python, TypeScript, SQL, MATLAB, C++ (Arduino) | Numerical Methods, Signal Processing, Differential Equations
Software | Data: React, React Native, Expo, Flask, PostgreSQL, Backtrader, pandas, NumPy, SciPy, TensorFlow
Cloud | Development Tools: AWS, Supabase, Render, Vercel, Resend | Git, GitHub, Jira, Confluence, RevenueCat
Design | Manufacturing: Siemens NX, SolidWorks, Fusion 360 CAD/CAM | ACE CNC Certified

EXPERIENCE & LEADERSHIP

Design and Mfg. Lab Teaching Assistant **Aug 2024 – Present**
Design and Manufacturing Lab, UF *Gainesville, FL*

- **Teach and run the lab** for groups of four students through a semester-long build of compressed-air engines, owning their hands-on instruction on lathes, mills, bandsaws, MIG welders, and sheet metal equipment from first cut to final assembly.
- Mentor each team through design, fabrication, and assembly; review engineering drawings and reports for Design for Manufacturability and Assembly (DFMA) and grade assignments with feedback that drives improvement across the semester.
- Train students on shop safety at the start of every semester and enforce safety protocols during lab sessions; CNC-machine engine components and replacement parts for shop equipment to keep projects and machines running.

IPT Power Plant & Mech. Systems Engineering Intern **Jan 2025 – May 2025**
Piper Aircraft Inc. *Vero Beach, FL*

- **Self-taught heat transfer to develop a delay-differential model of aircraft brake rotor temperature** from ~47,000 airspeed and pressure sensor readings; implemented a 4th-order Runge-Kutta solver and mini-batch stochastic gradient descent with numerical derivatives to minimize prediction error.
- Built a Python sensor-data pipeline with CSV ingestion, Savitzky-Golay smoothing, and Akima interpolation, as well as a weighted decision-table classifier that used airspeed, brake pressure, and altitude to identify brake rotor heating and cooling conditions for binning a large dataset.
- Developed three aircraft configurations in Microsoft Flight Simulator for VR-based pilot evaluation, configured flight dynamics in XML, and tuned pitch and roll response from pilot feedback; wrote a 14-page build guide for implementation on a custom-built VR flight simulator for future project aircraft.
- Built a Python tool to quantify how aircraft nose length affects pilot visibility, using CAD geometry and trigonometric analysis across 0 to 19 inches of extension to find the pitch-down needed to keep baseline visibility and inform configuration decisions.
- Reverse-engineered an aftermarket oleo strut using precise metrology techniques and tools, taking over 200 measurements over 2 weeks to fully model the device in Siemens NX; briefed a team of 5+ engineers on the mechanism of the strut to better understand how to design an in-house strut.

INVOLVEMENT

Knead, *Co-Founder and Chief Technology Officer* **Spring 2026 – Fall 2026**

- Architected and built the **Strategy Playground** in React and TypeScript, a visual trading-strategy builder where users drag Blockly blocks that a custom abstract syntax tree (AST) translator converts into Python, run one-click Backtrader backtests, and read results on TradingView charts with buy/sell markers, indicator overlays, Sharpe ratio, and max drawdown benchmarked against the S&P 500 (sample available on my website!).
- Built a working prototype version of each backend service, including the backtesting engine behind a **Flask API gateway**, polished each service once it worked at a basic level by delegating to our engineers, and partnered with the lead engineer to turn the system into a full **microservices backend**.
- Designed a live-trading architecture with per-strategy subprocesses, threaded market-data delivery, and a rate-limited order scheduler to keep broker credentials outside strategy code and respect API limits.
- Hired, led and mentored 5+ engineers, fully architecting the recruiting and technical interview internal process and successfully obtaining talented engineers … developed the pitch deck and business plan that won the **2026 GatorTank pitch competition** ($1,200).

Level, *Co-Founder and Chief Technology Officer* **Fall 2026 – Present**

- Built and launched a calisthenics app on the App Store with React Native & Expo plus TypeScript and a Supabase PostgreSQL backend, all based on a co-founder's training system proven in a 700+ member club. Created a custom beta tester management software and ran a 28-day, 30+ user beta test.

Bike Bros, *Partner and Chief Operations Officer* **Summer 2025 – Present**

- Built a network of 4+ mechanics to source and repair high-quality vintage bicycles for resale at affordable prices.
- Bought, repaired and re-sold 30+ bikes across the greater Gainesville area, running marketing campaigns on Facebook Marketplace and Instagram.

Classical Pianist and Accompanist **Jan 2023 – Present**

- Collaborate with UF faculty, vocalists, instrumentalists, and composers for recitals, juries, and concerts; direct weekly worship music at a local Church with 20+ regular attendees while constantly learning new repertoire and perform solo and chamber programs for audiences of up to 3,000.

PROJECTS

- Automated **Fully Stressed Design of 2D trusses** with **Python** and **ABAQUS** FEA, scripting the resize-and-reanalyze loop so each member converges to its allowable stress, running 100+ sizing iterations and saving 4+ hours of manual work.
- Earned the **ACE CNC certificate** by programming, CNC-machining, and assembling 5 air-engine components in Fusion 360 on a Haas VF-3 mill.
- Designed and machined a custom steel pen over 10+ iterations on a manual lathe, self-learning single-point threading.